Mystery Language Project Guidelines

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A. Introduction

Thai (ISO 639-3: tha), also called Siamese, Standard Thai, Central Thai, or Thaiklang, is a fully developed language of Tai-Kadai languages, spoken by 20,200,000 in Thailand (2000), 400,000 in Khorat (1984) and 4,704,000 ethnic Chinese in China (1984) (Lewis, 2009). Thai is just one of many languages and dialects belonging to the historical or proto-Tai family, a subgroup of Tai-Kadai languages, which is further divided into three branches: Northern Tai, Central Tai[1] and Southwestern Tai (Hudak, 2009). As belonging to the third branch, Thai is developed and standardized from the regional dialect of Bangkok and the surrounding provinces, and serves as an official national language of Kingdom of Thailand (Tingsabadh & Abramson, 1999).

Quite different from American English, which belongs to Indo-European languages, the Thai sound system is best described in a relationship to the syllable, the tone-bearing unit. As a tonal language, which excludes American English, Thai has five lexical tones. However, American English speakers are able to easily pronounce the consonants, as Thai and American English share a majority of the same consonants. And the vowels are also fairly isomorphic with American English vowels (Hartmann & Hudak, 2000). As will be presented here, Thai has a phonemic inventory of nine pairs of monophthongs, three diphthongs, twenty-one consonants, as well as five tones.

B. Consonants

1. Charts and Descriptions

Thai has seven voiceless stops, namely two bilabial stops [p, p^h], two alveolar stops [t, t^h], two velar stops [k, k^h], and one glottal stop [ʔ]. It also has two voiced

[1] Central Tai should be distinguished from Central Thai. Central Tai is a subgroup of Tai-Kadai languages, while Central Thai refers to Thai as another name.

stops [b, d]. Among these stops, [p^h, t^h, k^h] are aspirated while others unaspirated. Thai also has one voiced bilabial nasal [m], one voiceless labiodental fricative [f], one alveolar nasal [n], and one velar nasal [ŋ]. There is a voiced alveolar trill [r] in Thai. Besides, it has two voiceless fricatives as alveolar [s] and glottal [h], three voiced approximants as alveolar [l][2], palatal [j], and velar [w]. Particularly, there is a pair of post-alveolar affricates, unaspirated [tɕ] and aspirated [$tɕ^h$][3].

Table 1. Consonants chart

		Bilabial	Labio-dental	Alveolar	Post-alveolar	Palatal	Velar	Glottal
Stop	Unaspirated	p b		t d			k	ʔ
	Aspirated	p^h		t^h			k^h	
Nasal		m		n			ŋ	
Trill				r				
Fricative			f	s				h
Approximant				l		j	w	
Lateral Approximant				l				
Affricate	Unaspirated				tɕ			
	Aspirated				$tɕ^h$			

The left column is voiceless while the right is voiced.

Tingsabadh and Abramson's (1999) consonants chart of Thai is quite similar to ours, except the place for the manner of [l]. They put [l] in the row of lateral approximant, shown by an arrow and [l] in the chart above.

All the consonants of Thai may appear in initial position, but only [p, t, k, m, n, ŋ, j, w] are permitted to appear in final position. Also there exist consonant clusters that would occur in initial position, namely [pr, pl, p^hr, p^hl, tr, t^hr, kr, kl, kw, k^hr, k^hl, k^hw] (Hudak, 2009).

2. Discussion of Differences

Luckily, Thai and American English share a majority of the same consonants. I did not find it difficult for me to distinguish all the consonants in Thai. Particularly,

[2] In order to correspond to the original consonant chart, the description here is approximant [l].
[3] The description here is post-alveolar, corresponding to the original consonant chart.

thanks to my language background of Mandarin and Wu Chinese, it made me easily distinguish the pair of post-alveolar affricates, unaspirated [tɕ] and aspirated [tɕ^h] and the glottal stop [ʔ][4], which is also included in the consonants of Mandarin and Wu Chinese. It seems our original consonant chart is the same as Tingsabadh & Abramson's (1999).

However, I described [l] as an approximant while the authors describe it as a lateral approximant. It's the common knowledge that [l] should be a lateral approximant in any languages it might occur. I think it's due to my carelessness that I mistakenly put the alveolar lateral approximant [l] into the approximant row. Furthermore, there is one place that should be doubted. Both our original and Tingsabadh & Abramson's (1999) chart consider the place of articulation of [tɕ, tɕ^h] is post-alveolar. However, it may cause confusion between [tʃ] and [tɕ]. More precisely, now I prefer to consider [tɕ, tɕ^h] as respectively, unaspirated, aspirated voiceless alveolo-palatal affricates.

As stated before, the consonants of American English and Thai have a large overlap, more specifically, as they both share bilabial stops [p, b], alveolar stops [t, d], velar stop [k], nasals [m, n, ŋ], labiodental fricative [f], alveolar fricative [s], glottal fricative [h], palatal approximant [j], velar approximant [w] and alveolar lateral approximant [l]. It would be easy for recognizers to match the shared sounds between Thai and American English.

Besides, Thai has a series of sounds that American English does not have. Firstly, there is a group of voiceless aspirated stops [p^h, t^h, k^h] and affricate [tɕ^h] in Thai. For American English speakers, it seems that they may have no problem to pronounce the three stops [p^h, t^h, k^h] since [p, t, k] in English are aspirated in word-initial position or initial in a stressed syllable (Ladefoged, 1999). However, the articulation of these aspirated [p^h, t^h, k^h] in Thai is usually lax and is followed by fairly heavy aspiration (Harris, 1972), while in American English, the aspirated [p, t, k] are tense. Thus, even [p, t, k] could be aspirated in American English, it is quite possible that [p^h, t^h, k^h] in Thai would have longer VOT than aspirated [p, t, k] in American English.

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[4] Example words as 寄 (to send) [tɕi^{51}] and 气 (air) [tɕ^{h}i^{51}] in Mandarin Chinese, and 樱 (cherry) [ʔã^{55}] in Wu Chinese (Shanghaiese, accurately).

Also, American English speakers probably may use [tʃ] to substitute [tɕʰ], as these share the same manner of articulation and quite close places (post-alveolar and alveolo-palatal) of articulation.

Secondly, the glottal stop [ʔ] in Thai is absent in American English. Thus, the recognizers may probably just ignore the sound in that the release of the oral and glottal closure is usually simultaneous, so that the glottal release of [ʔ] is not heard (Harris, 1972).

Thirdly, the alveolar trill [r] in Thai is also absent in American English. Thus, the recognizers may probably use alveolar approximant [ɹ] or tap [ɾ], which often appears as an allophone, to refer to [r] instead, as they all share the same place of articulation, though having different manners of articulation.

C. Vowels

1. Charts and Descriptions

In the vowels chart below, it shows that there are nine vowels in Thai. However, actually, this language has eighteen monophthongs, which is made up of nine minimal pairs in vowel duration (long-short pairs): front unrounded high [i, iː], mid-high [e, eː], mid-low [ɛ, ɛː], back unrounded high [ɯ, ɯː], mid-high [ɤ, ɤː], low [ɑ, ɑː], and back rounded high [u, uː], mid-high [o, oː], mid-low [ɔ, ɔː]. Hudak (2009) points out that the nuclei of long vowels may be interpreted as two instances of the corresponding short vowel: [ii, ee, ɛɛ, ɯɯ, ɤɤ, ɑɑ, uu, oo, ɔɔ].

Also Thai has three diphthongs: [ɯɑ, uɑ, iɑ].

Table 2. Vowels chart (original)

	Front	Back
High	i	ɯ u
Mid-High	e	ɤ o
Mid-Low	ɛ	ɔ
Low		ɑ

The left column is unrounded while the right is rounded.

	Front	Back
High	i	ɯ u
Mid-High		
Mid-Low		
Low		ɑ

Three diphthongs in this language

Tingsabadh and Abramson's (1999) vowels charts of Thai are almost the same as ours, as shown below in Table 3. More detailed than ours, they present an inventory,

which includes gloss and transcriptions as below, to show all short, long and diphthong vowels.

Table 2. Vowels chart (Tingsabadh and Abramson, 1999)

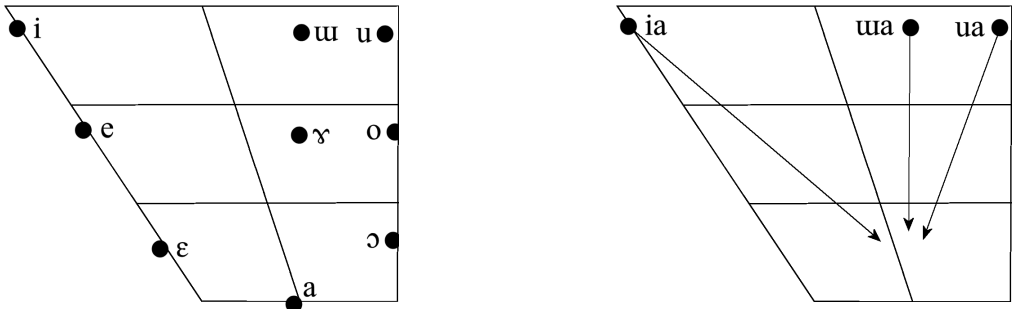

Chart reprinted from Tingsabadh and Abramson (1999).

i	krì t	'dagger'	iː	krìːt	'to cut'	ia	riān	'to study'
e	ʔēn	'ligament'	eː	ʔēːn	'to recline'	ɯa	rūan	'house'
ɛ	pʰɛ́ʔ	'goat'	ɛː	pʰɛ́ː	'to be defeated'	ua	rūan	'to be provocative'
a	fǎn	'to dream'	aː	fǎːn	'to slice'			
ɔ	klɔ̀ŋ	'box'	ɔː	klɔ̄ːŋ	'drum'			
o	kʰô̞n	'thick (soup)'	oː	kʰô̞ːn	'to fell (a tree)'			
u	sùt	'last, rearmost'	uː	sùːt	'to inhale'			
ɤ	ŋɤ̄n	'silver'	ɤː	dɤ̄ːn	'to walk'			
ɯ	kʰɯ̂n	'to go up'	ɯː	kʰlɯ̂ːn	'wave'			

2. Discussion of Differences

I was able to distinguish all the vowels in Thai and the most transcribed vowels are the same as Tingsabadh and Abramson's (1999). This is because as a Mandarin Chinese native speaker, I could easily tell the vowel [ɤ][5] which does not occur in American English. Furthermore, with the language knowledge of Japanese as the second foreign language, I could tell the vowel [ɯ][6] as well.

Only one difference between the two versions is that we choose [ɑ] in our original chart while they choose [a]. Tingsabadh and Abramson (1999) maintain that the vowel /a/ in unstressed position is likely to be somewhat raised in quality, and they choose /a/ to indicate both monophthongs and diphthongs. Hudak (2009)

[5] Example word as 鹅 (goose) [ɤ^{35}].

[6] Example word as うち (family) [ɯ tɕi].

also chooses /a/ to refer to the monophthong, with a description of the low back unrounded vowel, but he also use /ɑ/ to be combined with the initial vowels when describing the diphthongs. For me, I cannot accept the confusing usage of [a] and [ɑ], since it is contradictory to the IPA vowel chart. Thus, I prefer to use [ɑ] as a more precise transcription to refer to the low back unrounded vowel, which appears in both monophthongs and diphthongs of Thai vowels.

As the consonants, Thai vowels also share a portion of vowels, which appear in American English as well. They both share front unrounded high [i], mid-high [e], and mid-low [ɛ], back rounded high [u] and mid-high [o], and back low unrounded [ɑ]. Those vowels above may have no difficulties for American English speakers to pronounce.

However, there are still three vowels in Thai that American English does not have that back high unrounded [ɯ], back mid-high unrounded [ɤ], and back mid-low rounded [ɔ]. The first one is [ɯ]. The sound of [ɯ] may be not strange for American English speakers, as it sometimes does appear as an allophone of [ʊ] in some English dialects, like "good" [kɯd]. But, it does not help American English speakers to distinguish it as a phoneme. Thus, they may probably use [ʊ] to substitute [ɯ], as [ʊ] and [ɯ] both appear as back high vowels. Also they may use [u] as [u] and [ɯ] are tense vowels.

The second vowel is [ɤ], and the recognizers may probably choose [o] or [ɚ] instead, as these three vowels are quite close to each other. The third vowel is [ɔ]. Actually this one is familiar to Received Pronunciation English speakers, as it does exist in RP English vowel chart. American English speakers prefer to use [ɑ] instead of [ɔ]. Thus, we believe, the recognizers may probably, as they pronounce American English, choose [ɑ] instead of [ɔ].

D. Tones

1. Charts and Descriptions

Thai has five lexical tones: high falling (51), low falling (21), low falling-rising (214), high rising (45), and mid falling (32). Table 4 above displays a minimal set that illustrates the five lexical tones.

Table 4. Tonal contrasts

Gloss/File names	Transcription	Description	Pitch
I-fem	$k^h a^{51}$	high falling	51
'galangal'	$k^h a^{21}$	low falling	21
'leg'	$k^h a^{214}$	low falling-rising	214
'to engage in trade'	$k^h a^{45}$	high rising	45
'to get stuck'	$k^h a^{32}$	mid falling	32

Tingsabadh and Abramson (1999) use a different way of diacritics to indicate tones in Thai, as shown below. Five tones in Standard Thai are high /´/, mid /¯/, low /`/, rising /ˇ/, and falling /ˆ/, which include three level tones (high, mid, and low) and two contours (falling and rising).

kʰāː	'to get stuck'	kʰáː	'to engage in trade'
kʰàː	'galangal'	kʰǎː	'leg'
kʰâː	'I'		

Hudak (2009) raises a rough distribution of tones in different syllables. Based on tonal occurrences, syllables could be divided into three types. (1) Syllables ending in a long vowel, a semi-vowel, or a nasal. All five tones occur on these syllables; (2) syllables ending with a short vowel and a stop or no final may have either low or high tone; (3) syllables with a long vowel followed by a stop usually have low and falling tones.

2. Discussion of Differences

I used tone numbers to indicate Thai tones while Tingsabadh and Abramson (1999) choose to use diacritics. Thus, I make a table for comparison of two different versions of tone charts below.

According to the chart above, we could clearly find that our tone chart is quite different from that of Tingsabadh and Abramson (1999), as only one tone, the falling tone k^hâ, is the same. The high tone k^há is described as $k^h a^{45}$, the mid tone k^hā is

described as $k^h a^{32}$ and the low tone k^hà as $k^h a^{21}$ in our version. The reason for this difference is that the diacritics cannot distinguish the very slight contours 45, 32 and 21, and they prefer to transcribe them as level tones. Also the rising tone k^hǎ is transcribed as $k^h a^{214}$ in our version, as I believe there exists a slight falling before the rising.

Gloss/File names	Transcription	Pitch (number)	Pitch (diacritics)
I-fem	$k^h a^{51}$	51	k^hâ
'galangal'	$k^h a^{21}$	21	k^hà
'leg'	$k^h a^{214}$	214	k^hǎ
'to engage in trade'	$k^h a^{45}$	45	k^há
'to get stuck'	$k^h a^{32}$	32	k^hā

Generally speaking, using tone numbers may describe more specifically than using diacritics. I was able to distinguish all the tones because my mother tongue Mandarin Chinese is also a tonal language, and I am confident in my transcription of tones in Thai.

American English has stress, but is not a tonal language. Thus, for recognizers, Thai tones are brand-new to distinguish. However, I think certain stress and intonation in American English may also have some influence on their recognition of Thai tones. It would be quite interesting to delve the interaction between intonation and tones, but till now I have no idea about it.

E. Conclusion

Thai, as a major language in Tai-Kadai languages, is widely spoken around Thailand, Burma, Laos, Vietnam, and Yunnan Province of China. It has a phonemic inventory of nine pairs of monophthongs, three diphthongs, twenty-one consonants, as well as five tones.

To sum up, I really enjoyed the whole process of doing the mystery language project. This project is really well-organized to be carried out. The hardest part of

the project should be transcription while you have no idea what language you are listening to. However, that is also the most enjoyable part precisely because of the mystery. For me, I enjoyed the transcription of Thai tones. Because as a native speaker of Mandarin Chinese, I am quite familiar with tones, but I haven't got in touch with foreign tonal languages. I had a strong sense of déjà vu, but not really the same as I imagined, and sometimes I felt it even a little bit challenging to accurately transcribe the tones. Furthermore, in order to write the final report, it forced me reading a lot of relevant materials and using IPA repeatedly. The work really helps me to learn and consolidate phonetic and phonological knowledge.

I hope, if possible, the project might include more phonetic knowledge, as measuring VOT and pitch contours, in it, besides simple comparison between the target language and English.

F. Transcriptions

1. Consonants

In alphabetical order (as their file names/gloss show), as found in the sound file folder *Consonants*:

Gloss/File names	Transcription	Consonant
act	[kaː^{21}n]	[k]
alms	[tʰaː^{21}n]	[tʰ]
belligerent	[pʰaː^{21}n]	[pʰ]
birthmark	[paː^{32}n]	[p]
calloused	[taː^{51}n]	[t]
court	[saː^{214}n]	[s]
demon	[maː^{21}n]	[m]
dish	[tɕaː^{32}n]	[tɕ]
long time	[naː^{32}n]	[n]
million	[laː^{45}n]	[l]
saddle	[ʔaː^{32}n]	[ʔ]
sagging	[jaː^{32}n]	[j]

shaft	[kʰɑː^{32}n]	[kʰ]
shop	[rɑː^{45}n]	[r]
sugar palm	[dɑː^{32}n]	[d]
to bloom	[bɑː^{32}n]	[b]
to divide	[hɑː^{214}n]	[h]
to slice	[fɑː^{214}n]	[f]
to sow	[wɑː^{32}n]	[w]
trash	[tɕʰɑː^{32}n]	[tɕʰ]
work	[ŋɑː^{32}n]	[ŋ]

2. Vowels

2.1 Monophthongs

Arranged by short-long pairs to illustrate minimal pairs:

Short			Long		
Gloss File names	Transcription	Vowel	Gloss File names	Transcription	Vowel
box	[klɔ^{21}ŋ]	[ɔ]	*drum*	[klɔː^{21}ŋ]	[ɔː]
dagger	[kri^{21}]	[i]	*to cut*	[kriː^{21}]	[iː]
goat	[pʰɛ^{45}ʔ]	[ɛ]	*to be defeated*	[pʰɛː^{45}]	[ɛː]
last-rearmost	[su^{21}]	[u]	*to inhale*	[suː^{21}]	[uː]
ligament	[e^{32}n]	[e]	*to recline*	[eː^{32}n]	[eː]
silver	[ŋɤ^{32}n]	[ɤ]	*to walk*	[dɤː^{32}n]	[ɤː]
thick soup	[kʰo^{51}n]	[o]	*to fell a tree*	[kʰoː^{51}n]	[oː]
to dream	[fɑ^{214}n]	[ɑ]	*to slice*	[fɑː^{214}n]	[ɑː]
to go up	[kʰɯ^{51}n]	[ɯ]	*wave*	[kʰlɯː^{51}n]	[ɯː]

2.2 Diphthongs

Gloss/File names	Transcription	Vowel
house	[rɯɑn^{32}]	[ɯɑ]
to be provocative	[ruɑn^{32}]	[uɑ]
to study	[riɑn^{32}]	[iɑ]

3. Tones

In alphabetical order, as found in the sound file folder *Tones*:

Gloss/File names	Transcription	Description	Pitch
'galangal'	$k^h a^{21}$	low falling	21
I-fem	$k^h a^{51}$	high falling	51
'leg'	$k^h a^{214}$	low falling-rising	214
'to engage in trade'	$k^h a^{45}$	high rising	45
'to get stuck'	$k^h a^{32}$	mid falling	32

G. References

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